

MACHINE LEARNING

CORE CONCEPTS HANDBOOK

Model

A **Model** in Machine Learning is a mathematical function or algorithm that learns patterns from data and makes predictions.

Simple Meaning:

A model is a system that takes input (features) and produces output (prediction).

Example

Suppose we want to predict whether a ball is:

Tennis or Cricket

Features:

- Weight
- Surface Type

The model learns from previous examples and creates a decision rule like:

If weight < 80 AND surface = smooth → Tennis

Else → Cricket

This learned rule is the **model**.

Mathematical View

In Linear Regression:

$$y = mx + b$$

The equation itself becomes the model.

In Logistic Regression:

$$P(y=1) = 1 / (1 + e^{-(wx + b)})$$

The learned weights (w, b) define the model.

Important Point

Model = Learned Pattern

Not just code.

It is the trained mathematical representation.

Training

Training is the process of teaching the model using training data so that it learns patterns.

During training:

- Model sees input data (X)
- Model compares predicted output with actual output (y)
- Model adjusts internal parameters (weights)
- Error is minimized

Example

Training Dataset:

Weight	Surface	Type
60	Smooth	Tennis
90	Rough	Cricket

Model learns:

Smooth + Low Weight → Tennis

Rough + High Weight → Cricket

That learning process is **training**.

During training:

1. Model makes prediction

2. Calculate error (Loss Function)
3. Adjust parameters (Gradient Descent)
4. Repeat many times



Code Example

```
model.fit(X_train, y_train)
```

Here:

- `X_train` → features
- `y_train` → labels
- `fit()` → training process

Important Concept

Training data must contain:

- Inputs (`X`)
- Correct outputs (`y`)

Without correct outputs → cannot train supervised model.

Testing

Testing is the process of evaluating how well the trained model performs on unseen data.

The model has never seen this data before.

Testing checks:

Can model generalize to new data?

Example

After training ball classifier, we give new ball:

Weight = 70

Surface = Smooth

Model predicts: Tennis

If correct → good model

If wrong → model needs improvement



Code Example

```
predictions = model.predict(X_test)
accuracy = model.score(X_test, y_test)
```

Here:

- `X_test` → unseen data
- `score()` → testing accuracy

Difference Between Training and Testing

Term	Purpose	Data Used
Training	Learn patterns	Training dataset
Testing	Evaluate performance	Unseen dataset

Real-Life Analogy

Imagine preparing for exams.

Training:

You study from textbook and solve practice problems.

Testing:

Sent

Drafts

7

Purchases

12

More

Labels

+

Inbox

1,081

Starred

Snoozed

Sent

Drafts

7

Purchases

12

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+

If you only memorize examples → Overfitting
If you understand concepts → Good generalization



Important Related Terms

Parameters

Values learned during training.

Example:

- Slope in Linear Regression
- Weights in Neural Network

Hyperparameters

Values set before training.

Example:

- K in KNN
- Learning rate
- Max depth in Decision Tree